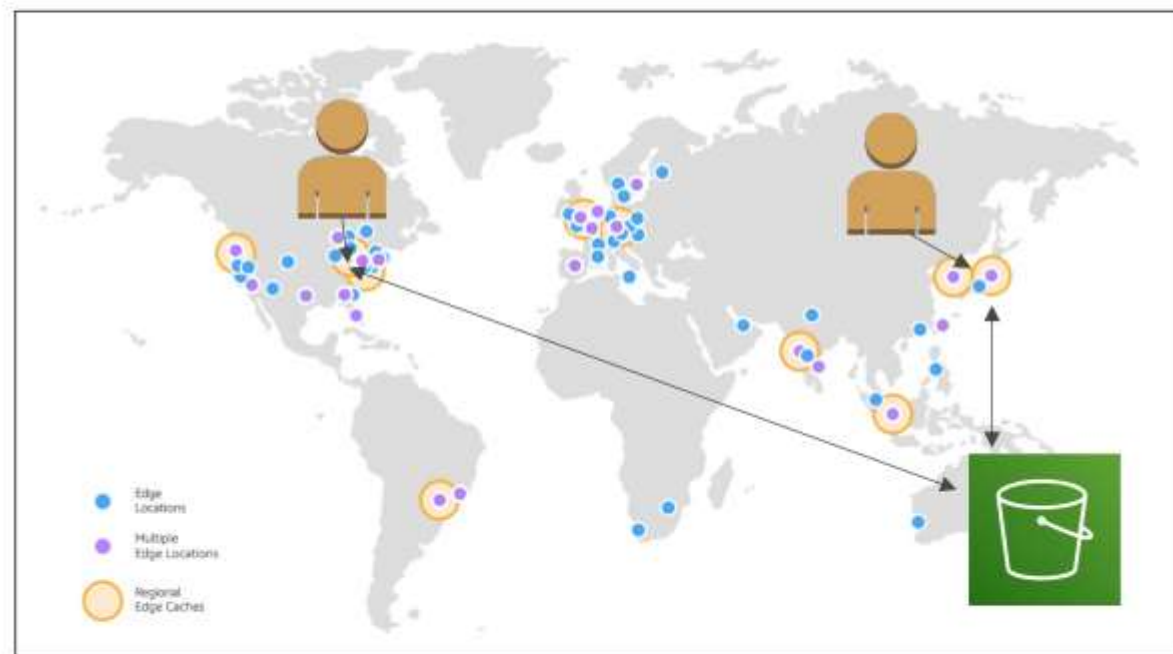


Amazon CloudFront



- Content Delivery Network (CDN)
- Improves read performance, content is cached at the edge
- Improves users experience
- Hundreds of Points of Presence globally (edge locations, caches)
- DDoS protection (because worldwide), integration with Shield, AWS Web Application Firewall



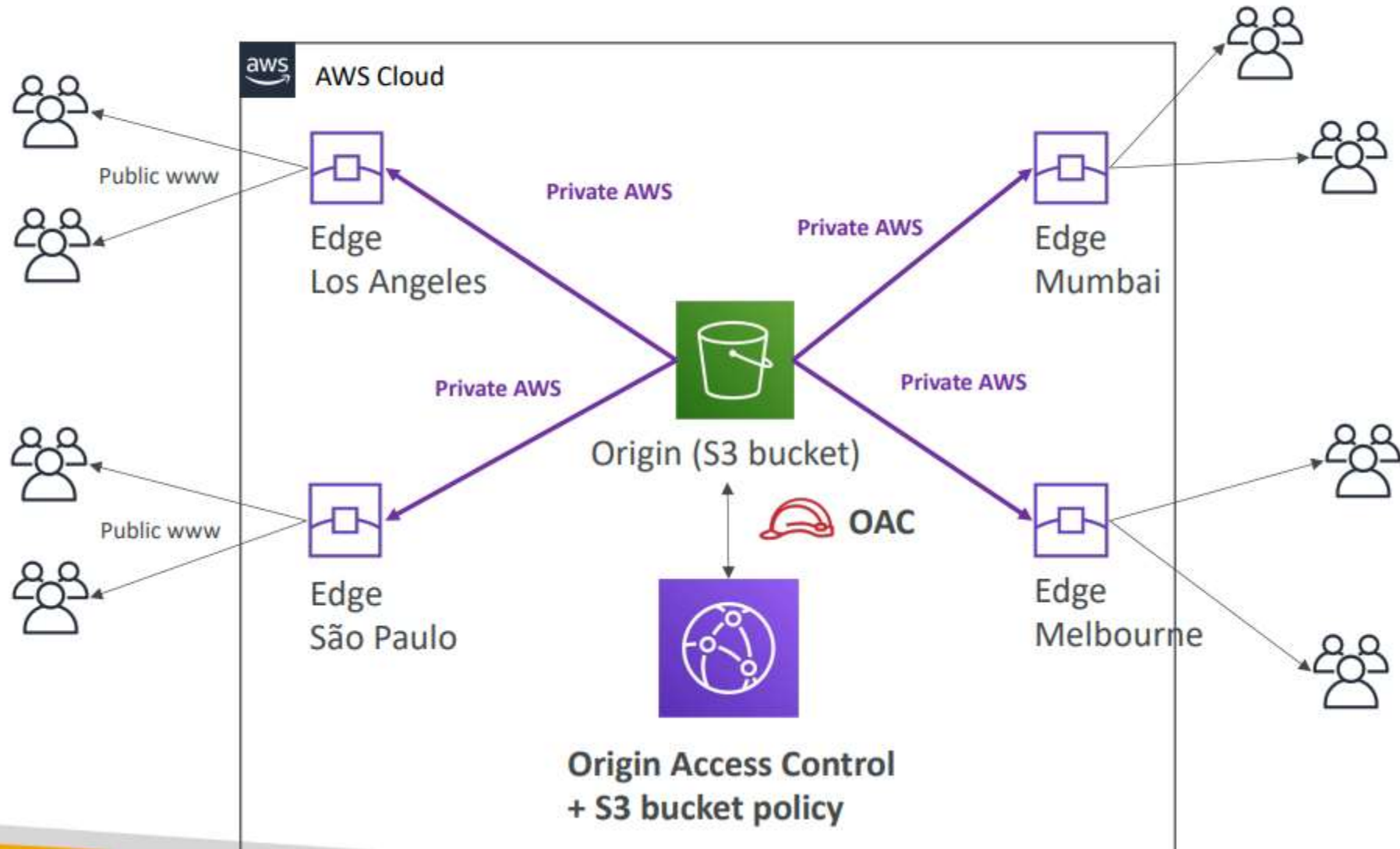
Source: <https://aws.amazon.com/cloudfront/features/?nc=sn&loc=2>

CloudFront – Origins

- **S3 bucket**
 - For distributing files and caching them at the edge
 - For uploading files to S3 through CloudFront
 - Secured using Origin Access Control (OAC)
- **VPC Origin**
 - For applications hosted in VPC private subnets
 - Private Application Load Balancer / Network Load Balancer / EC2 Instances
- **Custom Origin (HTTP)**
 - S3 website (must first enable the bucket as a static S3 website)
 - Any public HTTP backend you want (example: Public ALB)



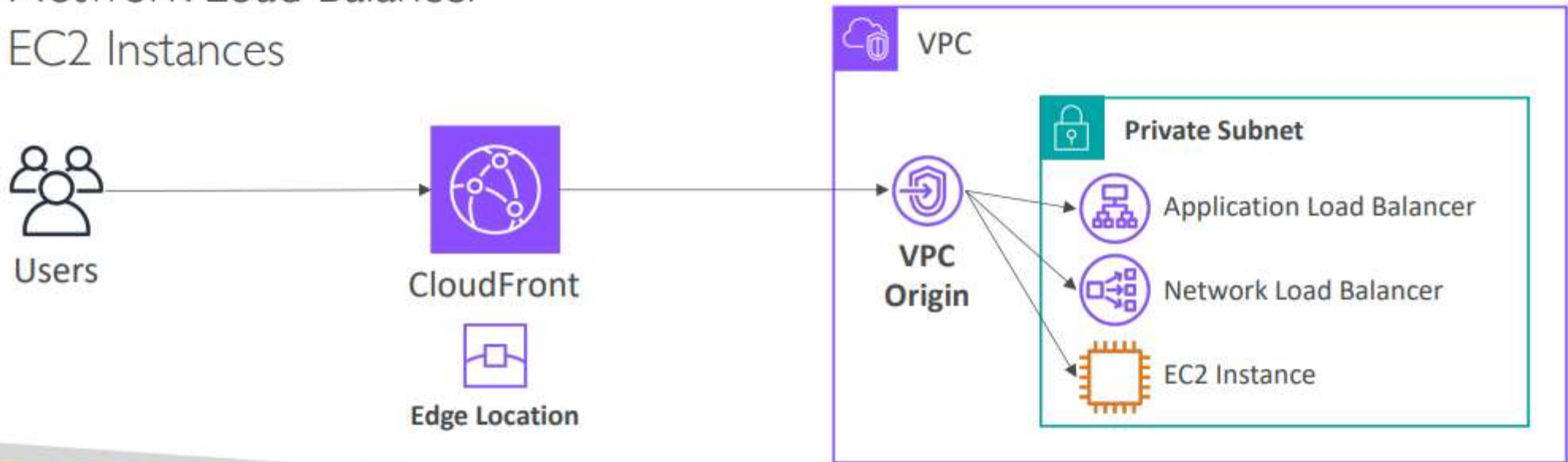
CloudFront – S3 as an Origin



CloudFront – ALB or EC2 as an origin

Using VPC Origins

- Allows you to deliver content from your applications hosted in your VPC private subnets (no need to expose them on the Internet)
- Deliver traffic to **private**:
 - Application Load Balancer
 - Network Load Balancer
 - EC2 Instances

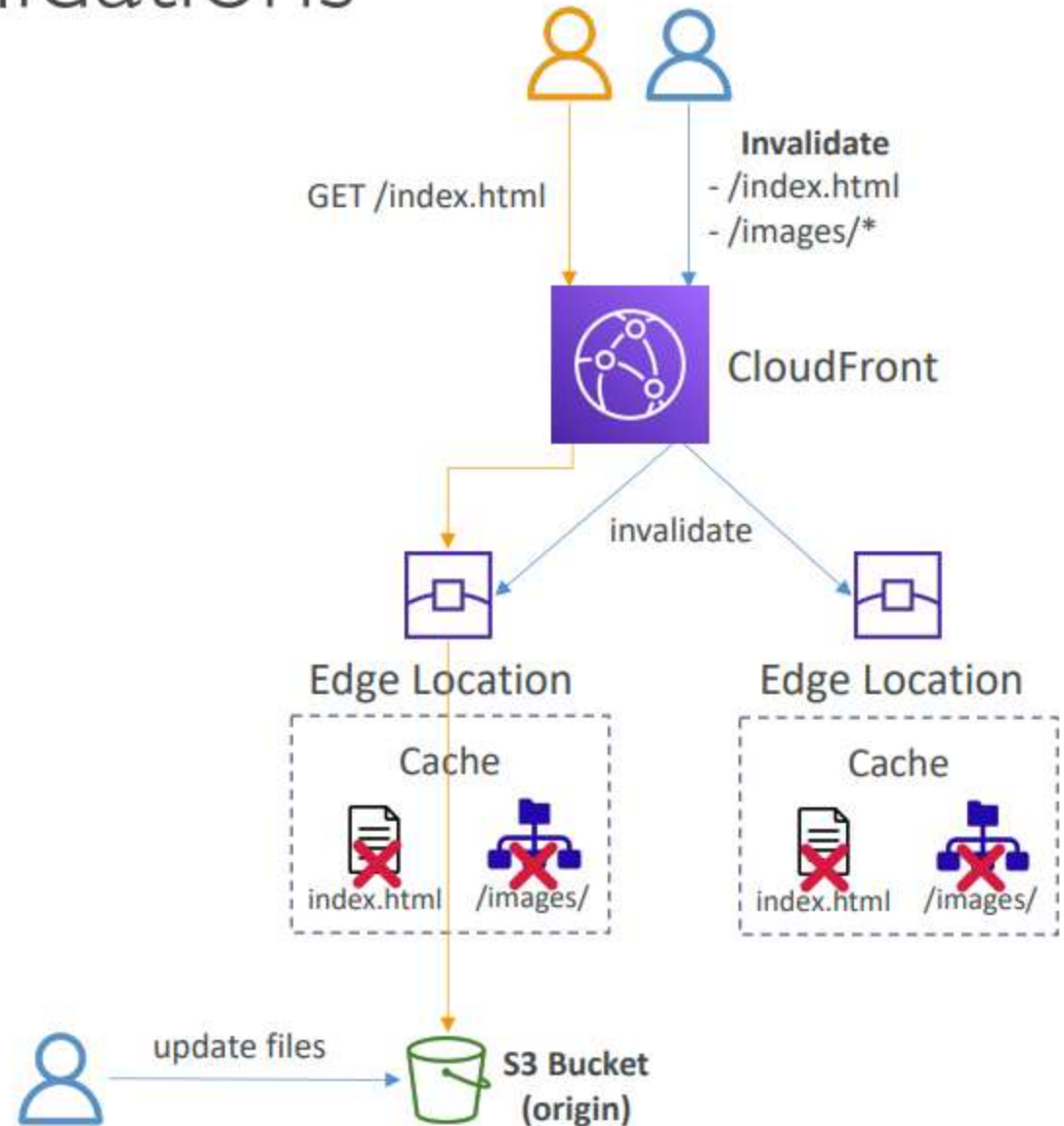


CloudFront Geo Restriction

- You can restrict who can access your distribution
 - **Allowlist:** Allow your users to access your content only if they're in one of the countries on a list of approved countries.
 - **Blocklist:** Prevent your users from accessing your content if they're in one of the countries on a list of banned countries.
- The “country” is determined using a 3rd party Geo-IP database
- Use case: Copyright Laws to control access to content

CloudFront – Cache Invalidations

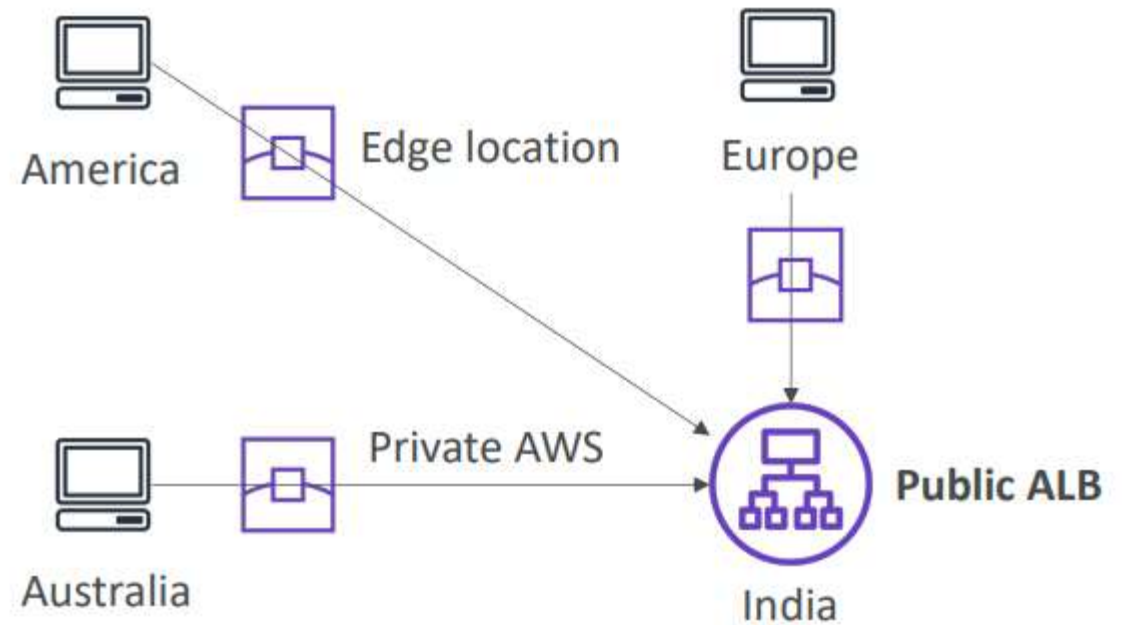
- In case you update the back-end origin, CloudFront doesn't know about it and will only get the refreshed content after the TTL has expired
- However, you can force an entire or partial cache refresh (thus bypassing the TTL) by performing a **CloudFront Invalidation**
- You can invalidate all files (*) or a special path (/images/*)



AWS Global Accelerator



- Leverage the AWS internal network to route to your application
- **2 Anycast IP** are created for your application
- The Anycast IP send traffic directly to Edge Locations
- The Edge locations send the traffic to your application



AWS Global Accelerator

- Works with Elastic IP, EC2 instances, ALB, NLB, public or private
- Consistent Performance
 - Intelligent routing to lowest latency and fast regional failover
 - No issue with client cache (because the IP doesn't change)
 - Internal AWS network
- Health Checks
 - Global Accelerator performs a health check of your applications
 - Helps make your application global (failover less than 1 minute for unhealthy)
 - Great for disaster recovery (thanks to the health checks)
- Security
 - only 2 external IP need to be whitelisted
 - DDoS protection thanks to AWS Shield

AWS Global Accelerator vs CloudFront

- They both use the AWS global network and its edge locations around the world
- Both services integrate with AWS Shield for DDoS protection.
- **CloudFront**
 - Improves performance for both cacheable content (such as images and videos)
 - Dynamic content (such as API acceleration and dynamic site delivery)
 - Content is served at the edge
- **Global Accelerator**
 - Improves performance for a wide range of applications over TCP or UDP
 - Proxying packets at the edge to applications running in one or more AWS Regions.
 - Good fit for non-HTTP use cases, such as gaming (UDP), IoT (MQTT), or Voice over IP
 - Good for HTTP use cases that require static IP addresses
 - Good for HTTP use cases that required deterministic, fast regional failover

AWS Snowball



- Highly-secure, portable devices to collect and process data at the edge, and migrate data into and out of AWS
- Helps migrate up to Petabytes of data

Device	Compute	Memory	Storage (SSD)
Snowball Edge Storage Optimized	104 vCPUs	416 GB	210 TB
Snowball Edge Compute Optimized	104 vCPUs	416 GB	28 TB



Snowball Edge

Data Migrations with Snowball

	Time to Transfer		
	100 Mbps	1Gbps	10Gbps
10 TB	12 days	30 hours	3 hours
100 TB	124 days	12 days	30 hours
1 PB	3 years	124 days	12 days

Challenges:

- Limited connectivity
- Limited bandwidth
- High network cost
- Shared bandwidth (can't maximize the line)
- Connection stability

AWS Snowball: offline devices to perform data migrations

If it takes more than a week to transfer over the network, use Snowball devices!

Solution Architecture: Snowball into Glacier

- Snowball cannot import to Glacier directly
- You must use Amazon S3 first, in combination with an S3 lifecycle policy



Amazon FSx for Windows (File Server)



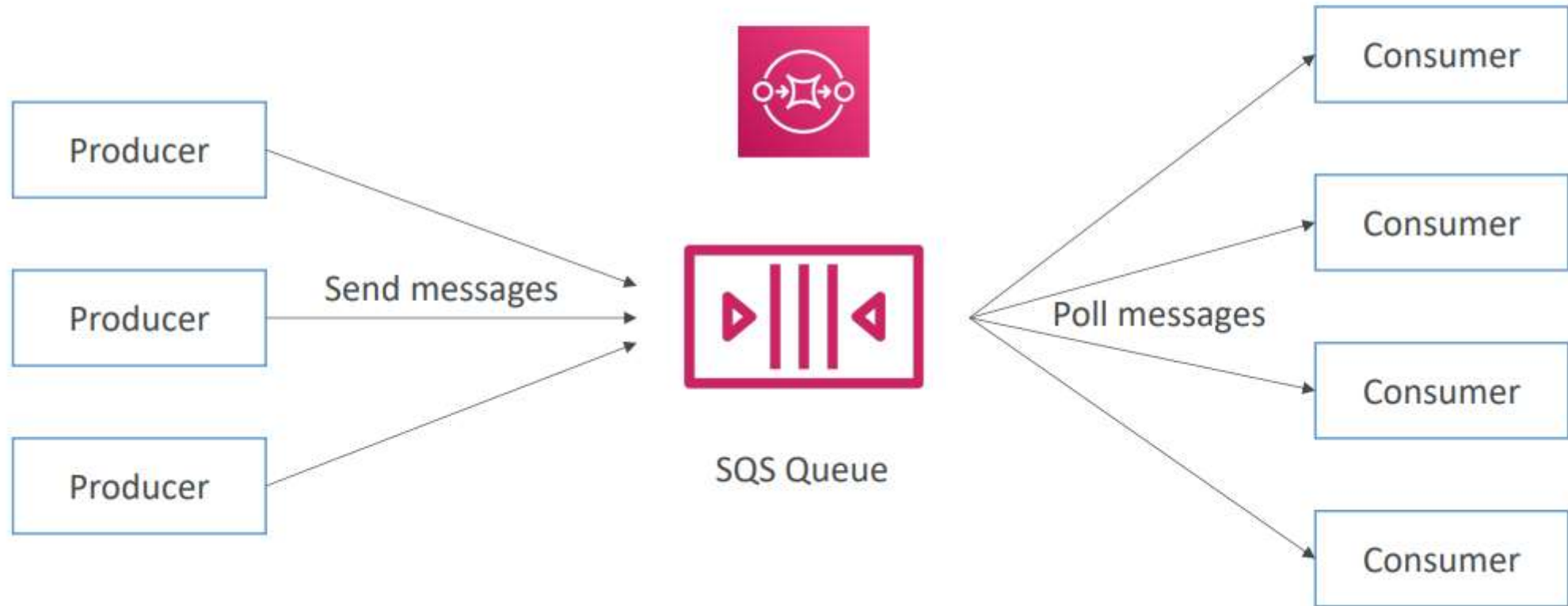
- **FSx for Windows** is a fully managed **Windows** file system share drive
- Supports SMB protocol & Windows NTFS
- Microsoft Active Directory integration, ACLs, user quotas
- **Can be mounted on Linux EC2 instances**
- Supports **Microsoft's Distributed File System (DFS) Namespaces** (group files across multiple FS)
- Scale up to 10s of GB/s, millions of IOPS, 100s PB of data
- Storage Options:
 - **SSD** – latency sensitive workloads (databases, media processing, data analytics, ...)
 - **HDD** – broad spectrum of workloads (home directory, CMS, ...)
- Can be accessed from your on-premises infrastructure (VPN or Direct Connect)
- Can be configured to be Multi-AZ (high availability)
- Data is backed-up daily to S3

Storage Comparison

- **S3:** Object Storage
- **S3 Glacier:** Object Archival
- **EBS volumes:** Network storage for one EC2 instance at a time
- **Instance Storage:** Physical storage for your EC2 instance (high IOPS)
- **EFS:** Network File System for Linux instances, POSIX filesystem
- **FSx for Windows:** Network File System for Windows servers
- **FSx for Lustre:** High Performance Computing Linux file system
- **FSx for NetApp ONTAP:** High OS Compatibility
- **FSx for OpenZFS:** Managed ZFS file system
- **Storage Gateway:** S3 & FSx File Gateway, Volume Gateway (cache & stored), Tape Gateway
- **Transfer Family:** FTP, FTPS, SFTP interface on top of Amazon S3 or Amazon EFS
- **DataSync:** Schedule data sync from on-premises to AWS, or AWS to AWS
- **Snowcone / Snowball / Snowmobile:** to move large amount of data to the cloud, physically
- **Database:** for specific workloads, usually with indexing and querying

Amazon SQS

What's a queue?



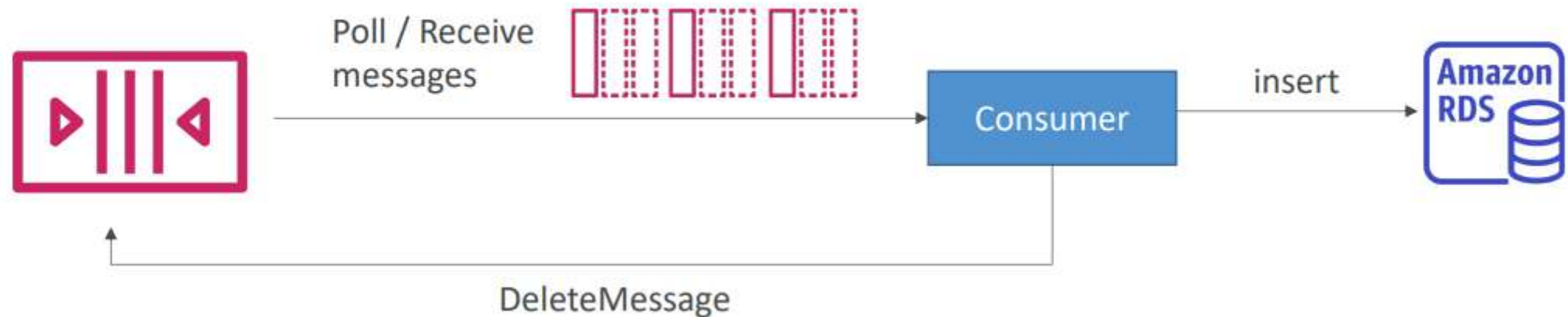
Amazon SQS – Standard Queue



- Oldest offering (over 10 years old)
- Fully managed service, used to **decouple applications**
- Attributes:
 - Unlimited throughput, unlimited number of messages in queue
 - Default retention of messages: 4 days, maximum of 14 days
 - Low latency (<10 ms on publish and receive)
 - Limitation of 1,024 KB per message sent
- Can have duplicate messages (at least once delivery, occasionally)
- Can have out of order messages (best effort ordering)

SQS – Consuming Messages

- Consumers (running on EC2 instances, servers, or AWS Lambda)...
- Poll SQS for messages (receive up to 10 messages at a time)
- Process the messages (example: insert the message into an RDS database)
- Delete the messages using the DeleteMessage API



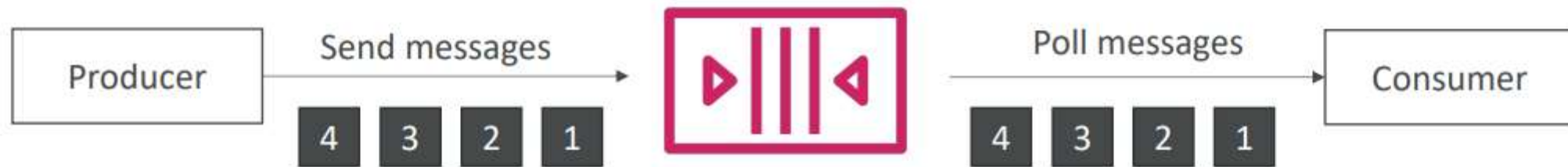
Amazon SQS - Security

- **Encryption:**
 - In-flight encryption using HTTPS API
 - At-rest encryption using KMS keys
 - Client-side encryption if the client wants to perform encryption/decryption itself
- **Access Controls:** IAM policies to regulate access to the SQS API
- **SQS Access Policies** (similar to S3 bucket policies)
 - Useful for cross-account access to SQS queues
 - Useful for allowing other services (SNS, S3...) to write to an SQS queue



Amazon SQS – FIFO Queue

- FIFO = First In First Out (ordering of messages in the queue)

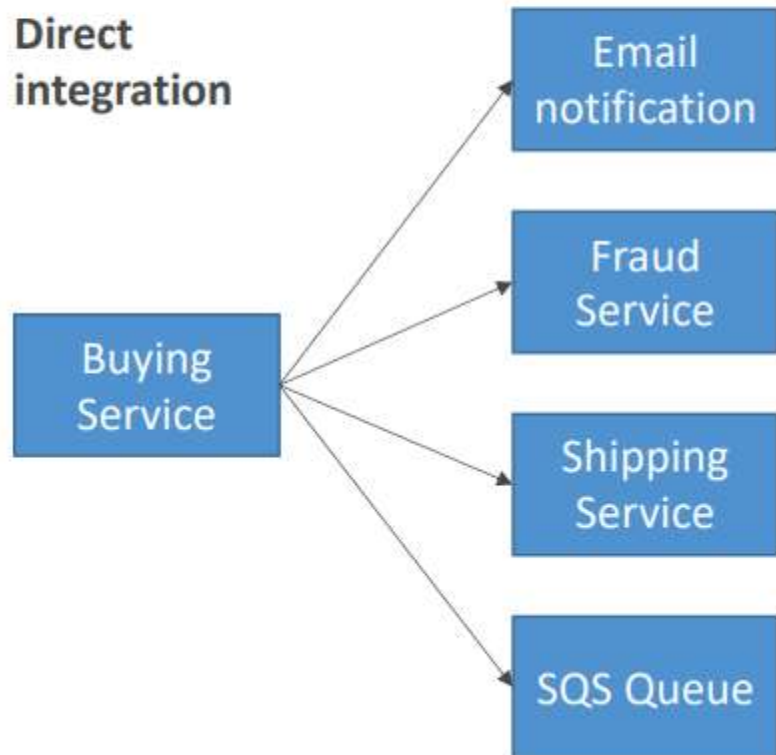


- Limited throughput: 300 msg/s without batching, 3000 msg/s with
- Exactly-once send capability (by removing duplicates using Deduplication ID)
- Messages are processed in order by the consumer
- **Ordering** by Message Group ID (all messages in the same group are ordered – mandatory parameter)

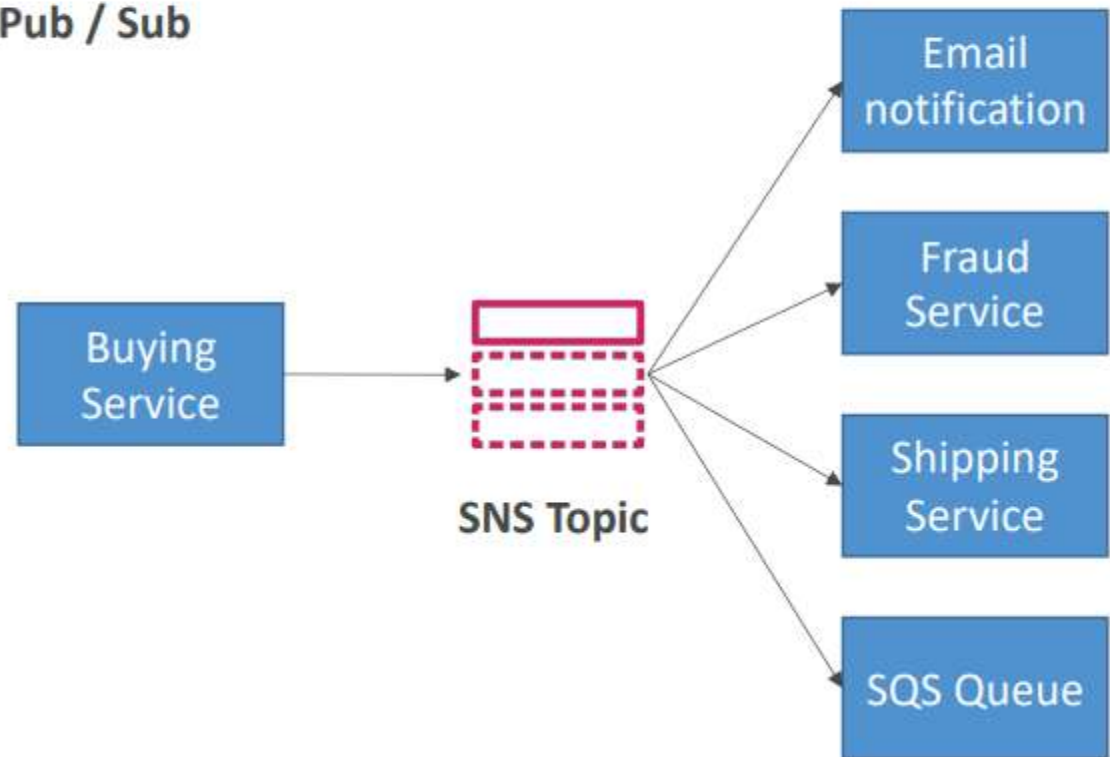
Amazon SNS

- What if you want to send one message to many receivers?

**Direct
integration**



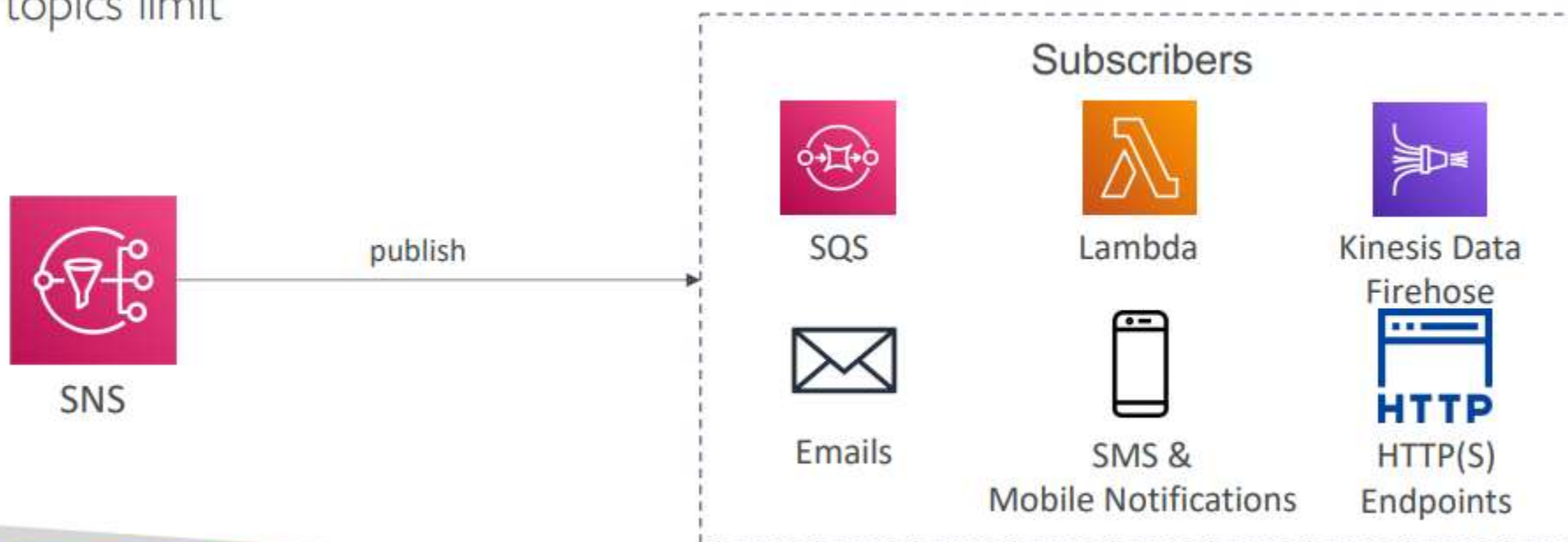
Pub / Sub



Amazon SNS



- The “event producer” only sends message to one SNS topic
- As many “event receivers” (subscriptions) as we want to listen to the SNS topic notifications
- Each subscriber to the topic will get all the messages (note: new feature to filter messages)
- Up to 12,500,000 subscriptions per topic
- 100,000 topics limit

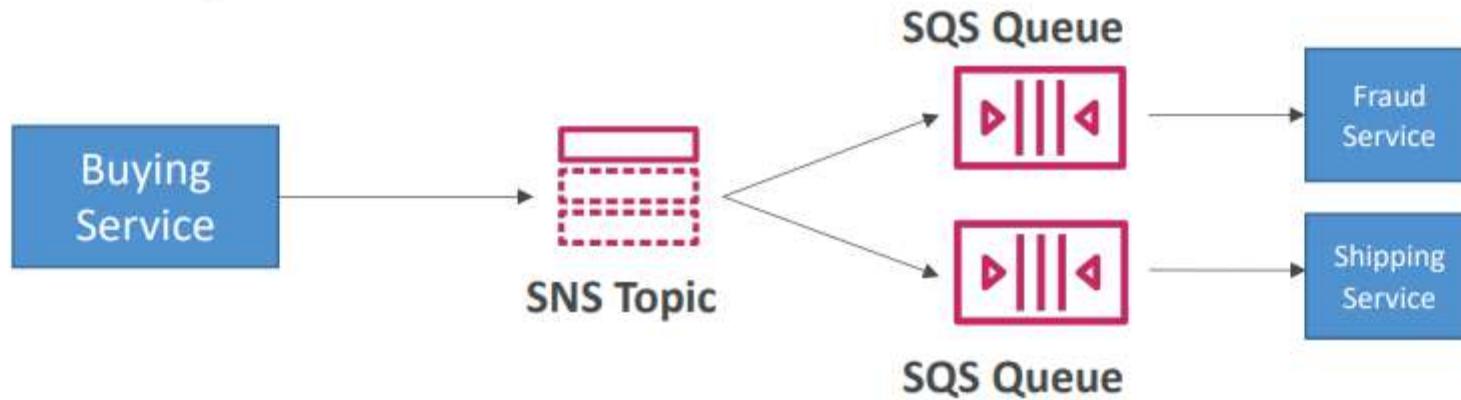


SNS integrates with a lot of AWS services

- Many AWS services can send data directly to SNS for notifications



SNS + SQS: Fan Out



- Push once in SNS, receive in all SQS queues that are subscribers
- Fully decoupled, no data loss
- SQS allows for: data persistence, delayed processing and retries of work
- Ability to add more SQS subscribers over time
- Make sure your SQS queue **access policy** allows for SNS to write
- Cross-Region Delivery: works with SQS Queues in other regions

Kinesis Data Streams



- Retention between up to 365 days
- Ability to reprocess (replay) data by consumers
- Data can't be deleted from Kinesis (until it expires)
- Data up to 10MiB (typical use case is lot of “small” **real-time data**)
- Data ordering guarantee for data with the same “Partition ID”
- At-rest KMS encryption, in-flight HTTPS encryption
- Kinesis Producer Library (KPL) to write an optimized producer application
- Kinesis Client Library (KCL) to write an optimized consumer application